Hashicorp

Documentation Infrastructure Audit Report

Prepared by VeriOps

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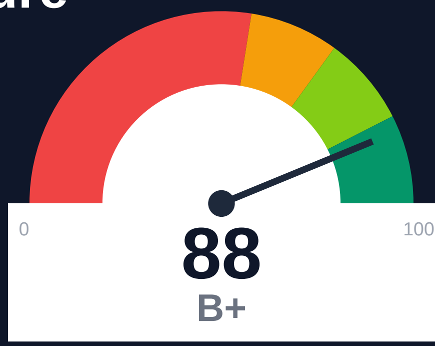


Risk: Low

Key Metrics

Pages scanned: 27659

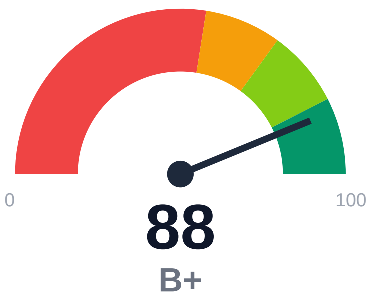
Report type: Premium Executive Audit

Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Crawl coverage is low: 48.96% (30000 URLs examined of 61271 in discovered scope). Run authenticated mode or in
- Broken internal links: 5024
- Link verification inconclusive for 839 URLs (auth/rate-limit/server block); these are excluded from broken-link counts.
- API usage-doc coverage is low: 47.5% (6037 API pages detected)

Executive Summary



Documentation audit of HashiCorp developer portal examined 30,000 URLs from a discovered scope of 61,271 pages, achieving 48.96% crawl coverage due to safety limits. The site contains 5,024 confirmed broken internal documentation links across 27,659 crawled pages. API reference coverage stands at 47.5% with 6,037 API pages detected. Freshness metadata is present on only 10.76% of pages, limiting change tracking visibility. Overall audit confidence is 78.12%, with crawl confidence reduced to 64.38% due to scope limitations.

External posture: SEO/GEO issue rate **2.8%**, public API coverage **47.5%**, broken links **5024**.

88 Audit Score	**B+** Grade	**Low** Risk Band
27659 Pages Scanned	**5024** Broken Links	**2.8%** SEO Issue Rate

Per-Site Breakdown

Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://developer.hashicorp.com/	27659	5024	47.5	100.0	2.8

Industry Benchmark Comparison

Benchmark	Score	Gap
Top performers (Stripe, Twilio)	**92+**	**-4**
Industry average (developer docs)	**85**	**+3**
Minimum acceptable	**70**	**+18**
Your score	**88**	**-**

Board-Level Metrics

API Coverage (Public) — **47.5%**

SEO/GEO Issue Rate (lower is better) — **2.8%**

Example Reliability (estimate) — **100.0%**

Freshness Visibility (last-updated metadata) — **10.8%**

Internal Metrics (from scorecard)

API Coverage (Internal) — **0.0%**

Example Reliability (Internal) — **100.0%**

Docs-Contract Drift — **0.0%**

Financial Exposure Model

Low Estimate	Base Estimate	High Estimate
$140,822 annual exposure	**$249,513** annual exposure	**$431,728** annual exposure
Remediation: **$2,090** Monthly loss: **$4,125** Opportunity: **$7,436**/mo	Remediation: **$5,445** Monthly loss: **$12,903** Opportunity: **$7,436**/mo	Remediation: **$8,800** Monthly loss: **$27,808** Opportunity: **$7,436**/mo

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-LAYERS	Missing required doc layers	HIGH	$1,782	$825
F-FRESHNESS	Documentation freshness debt	MEDIUM	$1,122	$550
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM	$2,257	$990
F-API-COVERAGE	Undocumented API operations	LOW	$2,904	$1,100
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW	$2,244	$935
F-DRIFT	Code/docs contract drift	LOW	$1,901	$660
F-TERMINOLOGY	Terminology inconsistency	LOW	$693	$385

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-LAYERS	Missing required doc layers	HIGH
F-FRESHNESS	Documentation freshness debt	MEDIUM
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM
F-API-COVERAGE	Undocumented API operations	LOW
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW

Recommended Actions

1. Fix 5,024 broken internal documentation links through automated link validation in CI/CD pipeline [impact: critical, effort: high]

2. Enable authenticated crawl mode to increase coverage from 48.96% and audit remaining 31,271 undiscovered pages [impact: critical, effort: medium]

3. Implement last-updated metadata across all pages to improve freshness tracking from 10.76% baseline [impact: high, effort: low]

4. Audit and document remaining 52.5% of API surface to achieve complete API reference coverage [impact: high, effort: high]

5. Investigate 839 unverified links with authentication/rate-limit issues to confirm actual status [impact: medium, effort: medium]

Expert Analysis: Strengths and Risks

Strengths

+ Example code reliability estimate at 100% indicates high quality of code samples
+ Multi-project topology successfully detected and analyzed across HashiCorp product suite
+ API pages successfully detected (6,037 pages) with no detection failures flagged
+ Zero repository navigation broken links, indicating clean separation of repo and docs infrastructure
+ SEO/geo issue rate low at 2.76%

Risks

- 5,024 broken internal documentation links create poor user experience and erode trust
- Crawl coverage at 48.96% means over half the documentation remains unaudited
- API reference coverage at 47.5% leaves significant API surface undocumented or unlinked
- 839 unverified links due to authentication or rate-limiting may hide additional broken links
- Only 10.76% freshness coverage prevents identification of stale or outdated content
- Low crawl confidence (64.38%) suggests potential authentication barriers or crawl traps affecting audit completeness
- 9,142 trap URLs skipped indicates complex URL patterns that may hide legitimate content issues

Limitations

* Crawl reached only 48.96% of discovered scope; findings do not represent full documentation estate and may miss critical issues in uncrawled sections
* 839 links could not be verified due to authentication, rate-limiting, or server blocks; actual broken link count may be higher
* External audit cannot verify accuracy of technical content, code functionality, or alignment with actual product API behavior
* Freshness analysis limited to 10.76% of pages with metadata; cannot assess staleness of remaining 89.24% of content
* Detection of 9,142 trap URLs may have excluded legitimate documentation pages requiring manual review

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	110.0
support_hourly_usd	50.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	12.0
avg_release_delay_hours	4.5
monthly_support_tickets	800.0
docs_related_ticket_share	0.22
avg_ticket_handling_minutes	24.0

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	HIGH

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.